

This document formalizes a **state-based rewriting logic** for an “operator word” calculus generated by:

- **Slides** $S_{ij}^\pm$
- **Blow up/down** $B_\pm, B_\pm^{-1}$
- **Hyperbolic stabilization/cancellation** H, H^{-1}

The calculus acts on configurations (Q, p) , where Q is an integral symmetric matrix and p is an integral vector (a characteristic/Pontryagin-refinement datum). A deterministic rewrite strategy produces a **canonical normal form** $\text{NF}(Q, p)$ that is invariant under the generated equivalences.

1. Configurations and semantics

1.1 Configuration space

A configuration is a pair

$$(Q, p), \text{ where } Q \in \text{Sym}_n(\mathbb{Z}), p \in \mathbb{Z}^n.$$

Interpretation: Q is a lattice pairing (e.g., an intersection form on a chosen basis), and p is extra structure (e.g., the spin characteristic element / Pontryagin refinement in that basis).

1.2 Congruence (“gauge”) action

A unimodular basis change $U \in GL_n(\mathbb{Z})$ acts by

$$(Q, p) \mapsto (U^T Q U, U^{-1} p).$$

Slides are elementary generators for this action.

2. Generator operators (raw moves)

For $i \neq j$ and $\varepsilon \in \{+1, -1\}$, let

$$U_{\{ij\}}^{\varepsilon} = I + \varepsilon E_{ij} \in GL_n(\mathbb{Z}),$$

where E_{ij} has a single 1 in position (i, j) .

Definition (Slide):

$$S_{\{ij\}}^{\varepsilon}(Q, p) = (U_{\{ij\}}^{\varepsilon})^T Q U_{\{ij\}}^{\varepsilon}, U_{\{ij\}}^{\varepsilon^{-1}} p.$$

Slides are total and invertible: $(S_{ij}^\varepsilon)^{-1} = S_{ij}^{-\varepsilon}$.

For $\sigma \in \{+1, -1\}$, define

$$B_{\sigma}(Q, p) = (Q \oplus [\sigma], p \oplus 0).$$

This is total.

Blow-down B_{σ}^{-1} is **partial**: it is admissible only if (Q, p) contains an orthogonal primitive σ -vector (precisely defined in §3.2).

Let $H_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Define

$$H(Q, p) = (Q \oplus H_2, p \oplus (0, 0)).$$

Cancellation H^{-1} is **partial**: it is admissible only if (Q, p) contains an orthogonal hyperbolic plane (precisely defined in §3.3).

3. Rewriting logic: explicit rules on configurations

We write rewrite rules as

$$\langle Q, p \rangle \rightarrow \langle Q', p' \rangle$$

possibly under an **admissibility predicate**.

3.1 Slide rule (always admissible)

(SLIDE)

$$\frac{\langle Q, p \rangle \rightarrow \langle U^T Q U, U^{-1} p \rangle}{\langle Q, p \rangle \rightarrow \langle U^T Q U, U^{-1} p \rangle}$$

where $U = U_{ij}(\epsilon) = I + \epsilon E_{ij}$.

3.2 Blow-up rule (always admissible)

(BLOWUP)

$$\frac{\langle Q, p \rangle \rightarrow \langle Q \oplus [\sigma], p \oplus 0 \rangle}{\langle Q, p \rangle \rightarrow \langle Q \oplus [\sigma], p \oplus 0 \rangle} \quad \sigma \in \{\pm 1\}.$$

We formalize blow-down as: find a unimodular U such that

$$U^T Q U = Q \oplus [\sigma].$$

Let π be projection onto the first $n - 1$ coordinates.

Predicate ***Split $_{\sigma}$ (Q)*****:** there exists primitive $v \in \mathbb{Z}^n$ and $U \in GL_n(\mathbb{Z})$ with $Ue_n = v$ and $U^TQU = Q' \oplus [\sigma]$.

(BLOWDOWN)

$$\frac{\mathrm{Split}_{\sigma}(Q) \wedge U^TQU = Q' \oplus [\sigma]}{Q, p \rightarrow Q', \pi(U^{-1}p)}$$

In rewriting-to-normal-form we do **not** allow arbitrary U ; we choose a *canonical* witness U (see §4.1).

3.3 Hyperbolic add rule (always admissible)

(HYPADD)

$$\frac{}{Q, p \rightarrow Q \oplus H_2, p \oplus (0,0)}$$

3.3' Hyperbolic cancel rule (admissible when an orthogonal hyperbolic plane splits)

Predicate ***HypSplit(Q)*****:** there exist primitive $u, v \in \mathbb{Z}^n$ and $U \in GL_n(\mathbb{Z})$ so that $Ue_{n-1} = u, Ue_n = v$, and

$$U^TQU = Q \oplus H_2.$$

Equivalently, in the original basis: $u^TQu = 0, v^TQv = 0, u^TQv = 1$, and the span splits off orthogonally.

Let π project away the last two coordinates.

(HYPCANCEL)

$$\frac{\mathrm{HypSplit}(Q) \wedge U^TQU = Q \oplus H_2}{Q, p \rightarrow Q', \pi(U^{-1}p)}$$

Again, normalization uses a canonical choice of (u, v, U) (see §4.1).

4. Deterministic rewrite-to-normal-form strategy

The central idea is: **words are not normalized syntactically; states are normalized deterministically.**

Define a deterministic normalization map

$$\mathrm{N}: (Q, p) \mapsto \mathrm{NF}(Q, p).$$

We obtain a *normalized step relation*:

$$(Q, p) \xrightarrow{\mathsf{N}} (g(Q, p))$$

for any generator g admissible at (Q, p) .

4.1 Canonical witness selection for partial reductions

To make N deterministic, every time a partial rule (BLOWDOWN/HYPCANCEL) is applicable, we choose the witness by a fixed minimization procedure.

Canonical blow-down choice

Among all primitive v that split off a $[\sigma]$, select the one minimizing the tuple

$$(|v|_2, \mathrm{lex}(v)).$$

Then among unimodular U with $Ue_n = v$ and $U^T Q U = Q' \oplus [\sigma]$, choose the U minimizing a deterministic tie-break key (e.g., lexicographic order of the last row, then of the full matrix entries).

Canonical hyperbolic cancellation choice

Among all primitive hyperbolic pairs (u, v) that split off H_2 , select the one minimizing

$$(\max(|u|_2, |v|_2), \mathrm{lex}(u), \mathrm{lex}(v)).$$

Then choose a canonical completing U as above.

Practical note: in plumbing testbeds, a bounded search over short primitive vectors/pairs typically finds these quickly.

Stage N1 — Stabilization reduction

1. While a canonical blow-down is admissible, apply (BLOWDOWN) using the canonical witness.
2. While a canonical hyperbolic cancellation is admissible, apply (HYPCANCEL) using the canonical witness.
3. Repeat until neither applies.

This strictly decreases rank whenever it fires, so it terminates.

Stage N2 — Global congruence normalization (real-place basis fix) Apply a deterministic lattice reduction procedure to fix the remaining congruence freedom, e.g.

- compute an LLL-reduced basis for the lattice with Gram matrix Q ,
- among all LLL-valid reduced bases, choose the one that lexicographically minimizes the coordinates of p ,

- fix a sign convention (first nonzero entry of p positive; then lex-min for remaining symmetries).

This yields a reproducible representative in the congruence class.

Stage N3 — Prime-local block normalization (optional but recommended) For each selected prime ℓ (and optionally a depth r):

1. Compute the ℓ -adic Jordan decomposition of $(Q, p) \otimes \mathbb{Z}_\ell$:

$$\mathbb{Z}_\ell \cong \bigoplus_{e \geq 0} \ell^e (Q_{\ell, e}, p_{\ell, e}).$$
2. If using a projector Π_{ℓ^r} , truncate to $0 \leq e < r$.
3. Sort blocks by a canonical key, e.g. $(e, \det \bmod (\mathbb{Z}_\ell^\times)^2, \text{Hasse/2-adic refinements}, \text{coordinates of } p_{\ell, e})$.

The result is $\text{NF}(Q, p)$. If you only need a stable normal form, you can stop at N2; N3 makes the prime-tier structure explicit.

5. Operator words, evaluation, and equivalence

5.1 Words

A word is a sequence of generator symbols

$w = g_m \cdots g_2 g_1$ subject to admissibility of partial inverses at runtime.

5.2 Normalized evaluation

Define $\text{EvalNF}(w; Q, p) := \text{NF}(\llbracket w \rrbracket(Q, p))$.

Two words are equivalent on input (Q, p) if they yield the same normal form.

5.3 “Resonance = 1” criterion (topological/move invariance)

Replace any scoring notion by strict equality of canonical outputs:

$\text{EvalNF}(w_1; Q, p) = \text{EvalNF}(w_2; Q, p)$.

This is the move-invariance condition.

This section makes (Q, p) **explicitly computable** from an 8D plumbing of disk bundles over S^4 whose boundary is a homotopy 7-sphere (Milnor-type).

6.1 Plumbing input

A plumbing description consists of:

- A finite graph $\Gamma = (V, E)$.
- For each vertex $v \in V$, an oriented D^4 -bundle over S^4 with structure group $SO(4)$, specified by **bundle parameters** $(a_v, b_v) \in \mathbb{Z} \oplus \mathbb{Z}$ (a standard $\pi_3(SO(4)) \cong \mathbb{Z} \oplus \mathbb{Z}$ coordinate choice).
- Each edge represents a plumbing along fiber/base disks with a chosen sign convention.

Let $n = |V|$ and choose the basis $\{x_v\}_{v \in V} \subset H^4(W; \mathbb{Z})$ given by the core S^4 of each bundle.

Define $Q \in \text{Sym}_n(\mathbb{Z})$ by:

- **Diagonal:** $Q_{vv} = e_v := a_v + b_v \setminus \setminus \text{Euler number in this convention}$.
- **Off-diagonal:** for $v \neq w$, $Q_{vw} = \sum_{\text{edges } (v,w)} \epsilon_{vw}$, where $\epsilon_{vw} \in \{\pm 1\}$ is the plumbing sign (often all +1 in standard Milnor plumbings).

This is the usual “diagonal = self-intersection, edges = intersection” plumbing matrix pattern, now in degree 4 for an 8-manifold.

In the same basis $\{x_v\}$, define a vector $p \in \mathbb{Z}^n$ by assigning each component from the vertex bundle parameters:

$(p)_v := a_v - b_v \setminus \setminus \text{a standard choice compatible with } p_1 \setminus \setminus \text{conventions up to overall factors}$.

Spin constraint / admissibility: to be in the spin setting, require the appropriate mod 2 parity condition on the vertex data (equivalently, the induced $w_2(W) = 0$); in many plumbing conventions this forces each Euler number e_v to be even.

If your convention uses p_1 directly, you can set $(p)_v$ proportional to the local p_1 -coefficient at the vertex; the rewrite calculus is unchanged (it only needs an integral vector transforming as $p \mapsto U^{-1}p$).

Once (Q, p) is formed for the coboundary W^8 , define:

- Signature $\sigma(W) = \text{sign}(Q)$.
- Quadratic value (over $\mathbb{Q}$) $p_W^2 := p^T Q^{-1} p \in \mathbb{Q}$.

Then a boundary invariant (for homotopy 7-spheres in the relevant class) is built from

$$\mu \equiv \frac{p_W^2 - \sigma(W)}{224} \pmod{1}$$

(up to convention choices). This makes the 2- and 7-primary structure explicit via the denominator.

6.5 Admissibility tests in plumbing cases (explicit)

In plumbing-derived Q , admissibility for cancellations often becomes concrete:

Blow-down test ($**\pm 1$ *****-sphere split): search for primitive v with

$$v^T Q v = \pm 1.$$

If found, attempt to complete v to a unimodular basis and check if the last basis vector is orthogonal to the others (i.e., off-diagonal last row/column can be eliminated by slides).

Hyperbolic split test: search for primitive u, v with

$$u^T Q u = 0, \quad v^T Q v = 0, \quad u^T Q v = 1,$$

and then verify orthogonal split by completing (u, v) to a unimodular basis and reducing cross-terms via slides.

These tests are exactly what Stage N1 requires.

7. Compact “rewriting logic” summary

7.1 Rule schema list

- (SLIDE) $\langle Q, p \rangle \Rightarrow \langle U^T Q U, U^{-1} p \rangle$ for $U = I \pm E_{ij}$.
- (BLOWUP) $\langle Q, p \rangle \Rightarrow \langle Q \oplus [\sigma], p \oplus 0 \rangle$.
- (BLOWDOWN) $\langle Q, p \rangle \Rightarrow \langle Q', \pi(U^{-1} p) \rangle$ if $U^T Q U = Q' \oplus [\sigma]$ (canonical witness).
- (HYPADD) $\langle Q, p \rangle \Rightarrow \langle Q \oplus H_2, p \oplus (0, 0) \rangle$.
- (HYPCANCEL) $\langle Q, p \rangle \Rightarrow \langle Q', \pi(U^{-1} p) \rangle$ if $U^T Q U = Q' \oplus H_2$ (canonical witness).

7.2 Deterministic strategy

Define $\text{NF}(Q, p) = \text{N}(Q, p)$ by:

1. Apply canonical (BLOWDOWN)/(HYPCANCEL) until stuck.
2. Apply deterministic congruence normalization (LLL + lex tie-break on p).
3. (Optional) Apply prime-local block normalization (Jordan decomposition + sorted blocks), including truncations Π_{ℓ^r} when desired.

8. Suggested “minimum viable experiment” specification (Milnor 7-spheres)

Input: a plumbing graph Γ with vertex parameters (a_v, b_v) and edge signs.

Compute:

1. (Q, p) via §6.2–§6.3.
2. $\text{NF}(Q, p)$ via §4.2.
3. Prime-tier fingerprints via Stage N3 (especially $\ell = 2$ and $\ell = 7$).
4. Scalar μ via §6.4, and its 2- and 7-primary projections.

Validate invariance: randomly apply sequences of slides (and optional stabilizations followed by normalization) and confirm $\text{NF}(Q, p)$ is unchanged.

9. Notes on conventions

- The precise mapping from vertex parameters (a_v, b_v) to $(e_v, (p)_v)$ depends on orientation/normalization conventions for $SO(4)$ -bundles over S^4 . The rewrite calculus itself is invariant under any consistent choice, as long as:
- Q is integral symmetric and transforms by congruence under slides,
- p transforms as $p \mapsto U^{-1}p$, and
- stabilizations append zeros to p .
- If you adopt a different convention, record it once and keep it fixed; $\text{NF}(Q, p)$ remains well-defined for that conventioer

References

1. **[MOC25]** *Multiplicity Operator Calculus (MOC)*, internal note/manuscript, Nov 5, 2025 (unpublished).
2. **[PMDM25]** *PMDM: Exotic Spheres direction*, internal note/manuscript (unpublished).
3. **[CN19]** D. Crowley, J. Nordström, “The classification of 2-connected 7-manifolds,” *Proc. London Math. Soc.* (3) **119**(1) (2019), 1–54. DOI: 10.1112/plms.12222. [Diarmuid Crowley+1](#)
4. **[CE03]** D. Crowley, C. M. Escher, “A classification of S^3S^3 -bundles over S^4S^4 ,” *Differential Geometry and its Applications* **18**(3) (2003), 363–380. DOI: 10.1016/S0926-2245(03)00012-3. [ScienceDirect](#)
5. **[DW59]** A. Dold, H. Whitney, “Classification of oriented sphere bundles over a 4-complex,” *Annals of Mathematics* (2) **69**(3) (1959), 667–677. DOI: 10.2307/1970030. [American Mathematical Society+1](#)
6. **[EK62]** J. Eells, Jr., N. H. Kuiper, “An invariant for certain smooth manifolds,” *Annali di Matematica Pura ed Applicata* (4) **60** (1962), 93–110. DOI: 10.1007/BF02412768. [Springer+1](#)

7. **[GS99]** R. E. Gompf, A. I. Stipsicz, *4-Manifolds and Kirby Calculus*, Graduate Studies in Mathematics 20, AMS, 1999. [Google Books+1](#)
8. **[Kir78]** R. Kirby, "A calculus for framed links in $S^3 \times S^3$," *Inventiones Mathematicae* **45** (1978), 35–56. DOI: 10.1007/BF01406222. [Springer+1](#)
9. **[KM63]** M. A. Kervaire, J. W. Milnor, "Groups of homotopy spheres. I," *Annals of Mathematics* (2) **77**(3) (1963), 504–537. [Uni-Math+1](#)
10. **[LLL82]** A. K. Lenstra, H. W. Lenstra Jr., L. Lovász, "Factoring polynomials with rational coefficients," *Mathematische Annalen* **261** (1982), 515–534. DOI: 10.1007/BF01457454. [Springer+1](#)
11. **[Mil56]** J. Milnor, "On manifolds homeomorphic to the 7-sphere," *Annals of Mathematics* (2) **64**(2) (1956), 399–405. [Department of Mathematics+1](#)
12. **[OM63]** O. T. O'Meara, *Introduction to Quadratic Forms*, Springer, 1963. [Semantic Scholar+1](#)
13. **[Ser73]** J.-P. Serre, *A Course in Arithmetic*, Springer, 1973. [Semantic Scholar+1](#)
14. **[CS99]** J. H. Conway, N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Springer, 1999. [Unina2+1](#)
15. **[KS77]** R. C. Kirby, L. C. Siebenmann, *Foundational Essays on Topological Manifolds, Smoothings, and Triangulations*, Princeton Univ. Press, 1977. [Department of Mathematics+1](#)
16. **[Wal62]** C. T. C. Wall, "Classification of $(n-1)(n-1)(n-1)$ -connected $2n$ -manifolds," *Annals of Mathematics* (2) **75** (1962), 163–189. [JSTOR+1](#)

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